Thomas Puskas

KUID: 2025952293

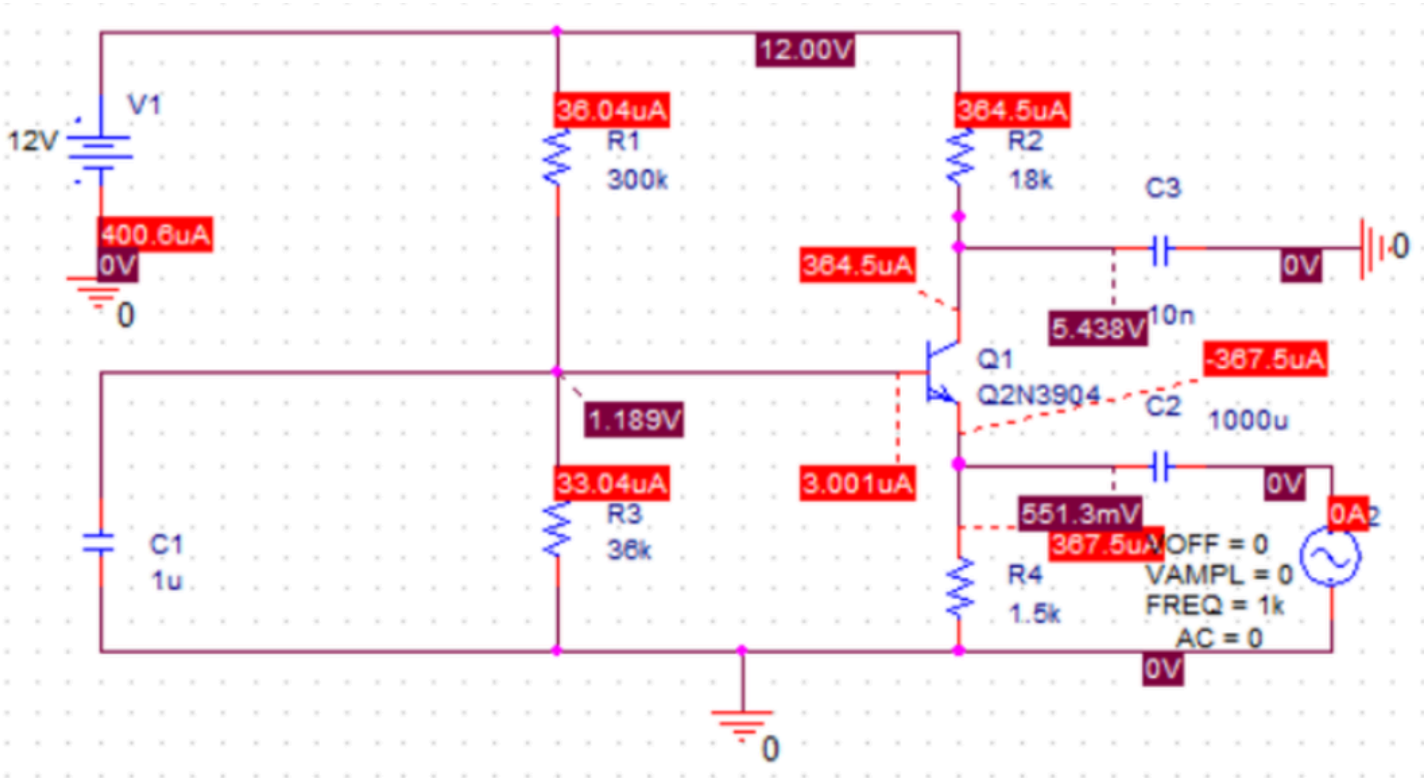
Experiment 8: BJT Common Base Amplifier, Emitter Follower

Objective:

The objective of this experiment is to practice and understand BJT common base amplifier and the emitter follower. We measure the gain and impedance of the BJT common base amplifier and emitters then compare the common base amplifier and emitter follower to each other at the points of gain and impedance.

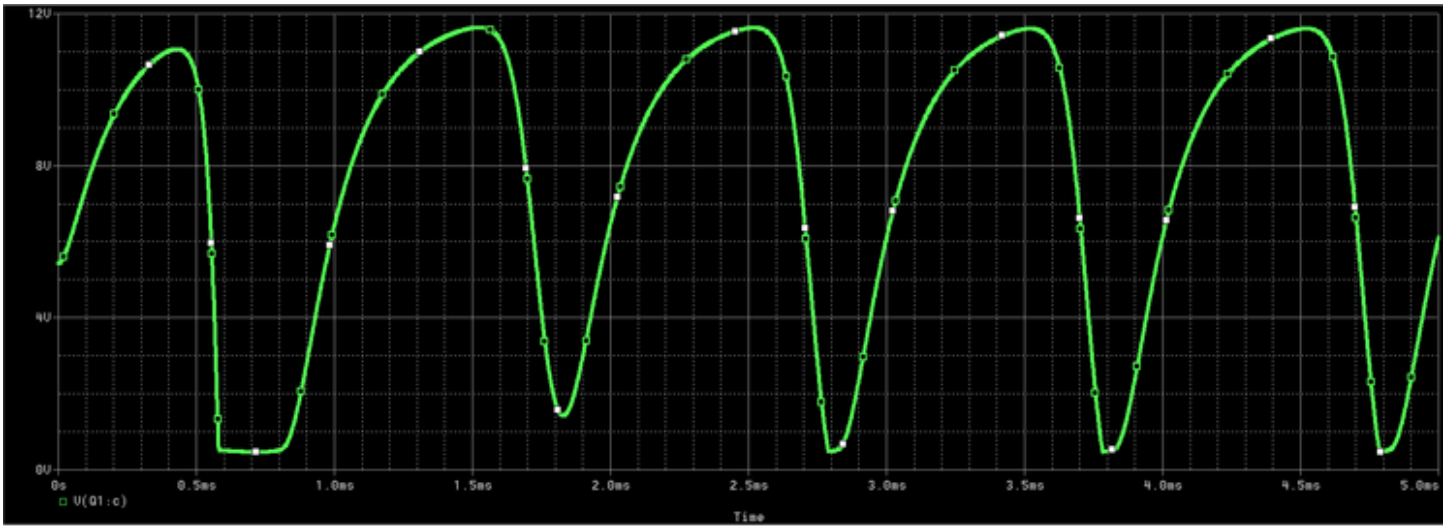
Pre-Lab:

Experiment 8.1: Calculate the parameters and find the operation point for given DC biased CB.

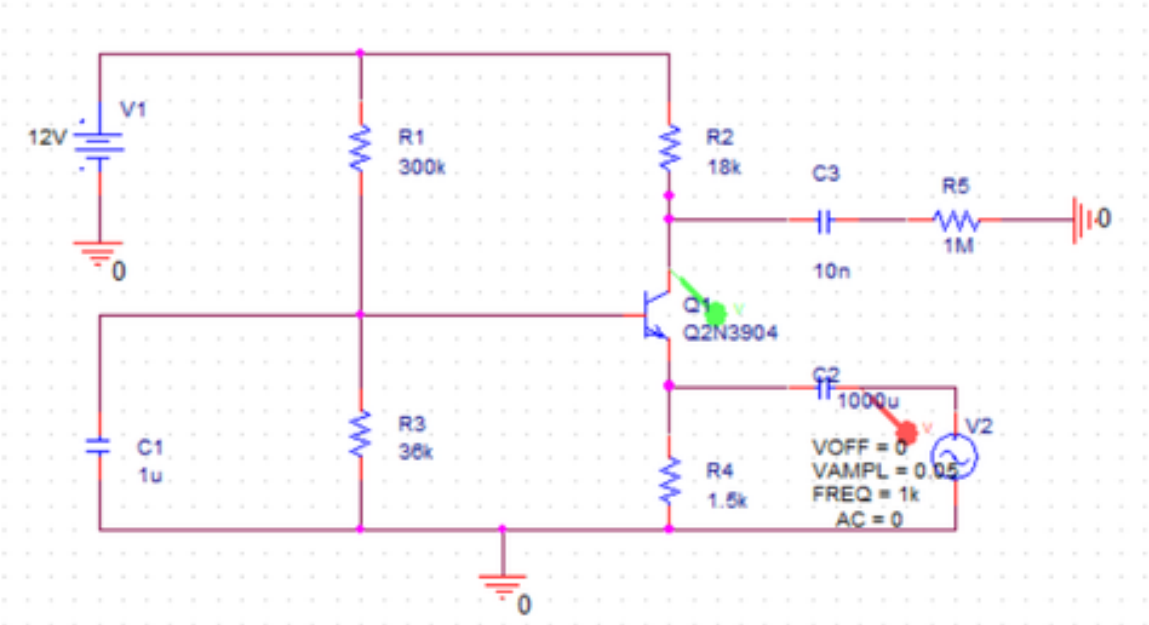


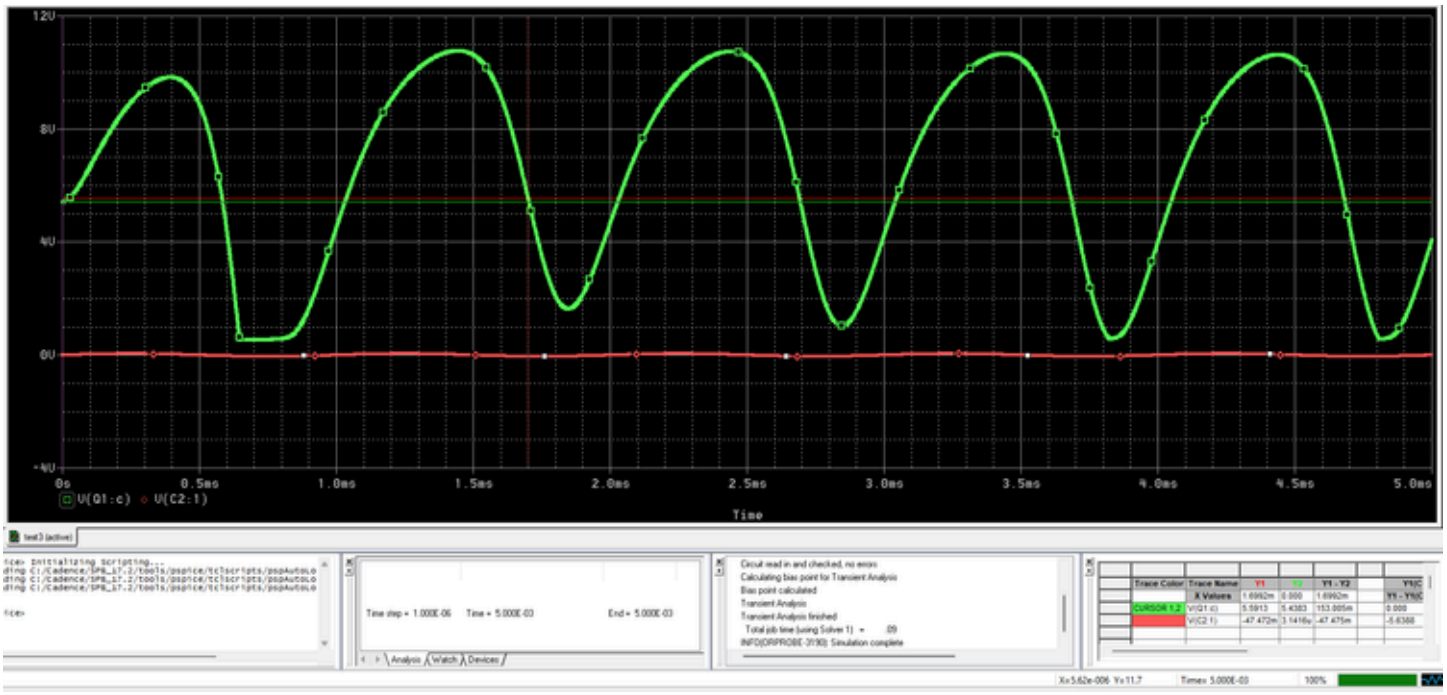
Common Base Amplifier					
V_B [V]	V_C [V]	V_E [mV]	I_B [µA]	I_C [µA]	I_E [µA]
1.189	5.438	551.3	3.001	364.5	367.5

Common Base Amplifier						
α	β	g_m [S]	r_π [Ω]	A_I	A_V (g_mR_C)	A_V (V_{out}/V_{in})
0.992	121.5	0.0141	8630	1.008	254	

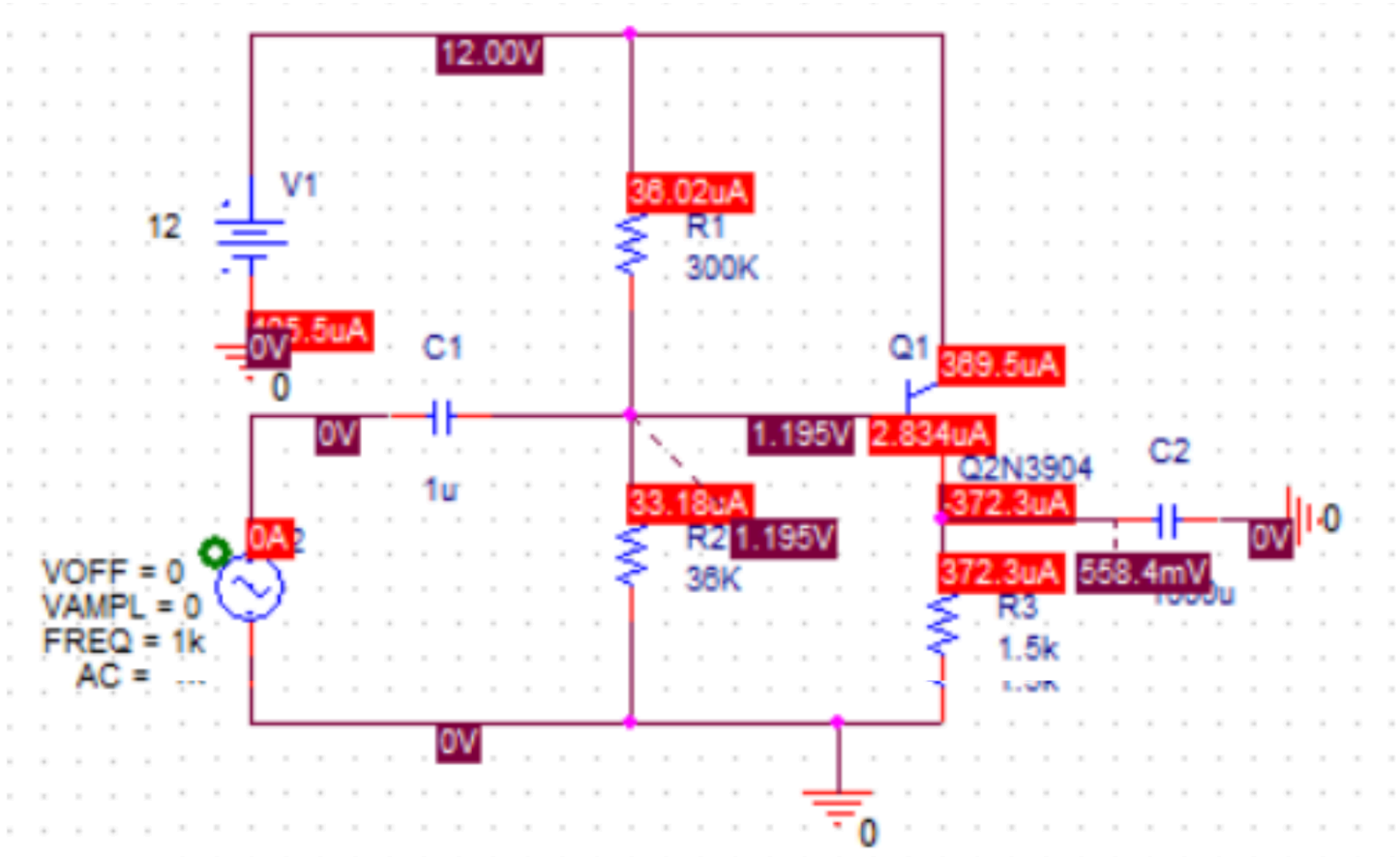


Experiment 8.2: Output Impedance of CB amplifier.



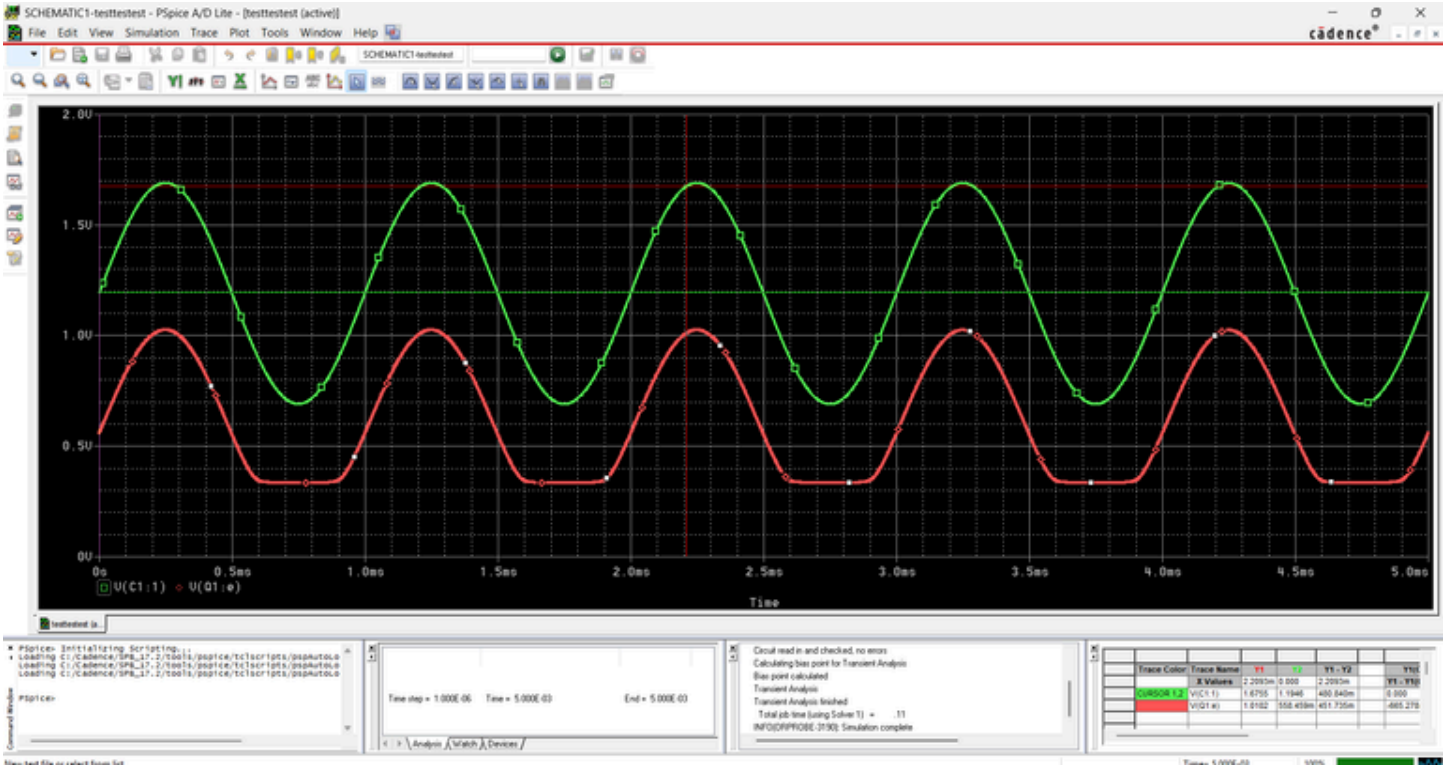


Experiment 8.3: Calculate the parameters and Output Impedance of EF



Common Base Amplifier					
V_B [V]	V_C [V]	V_E [mV]	I_B [µA]	I_C [µA]	I_E [µA]
1.195	12	558.4	2.834	369.5	372.3

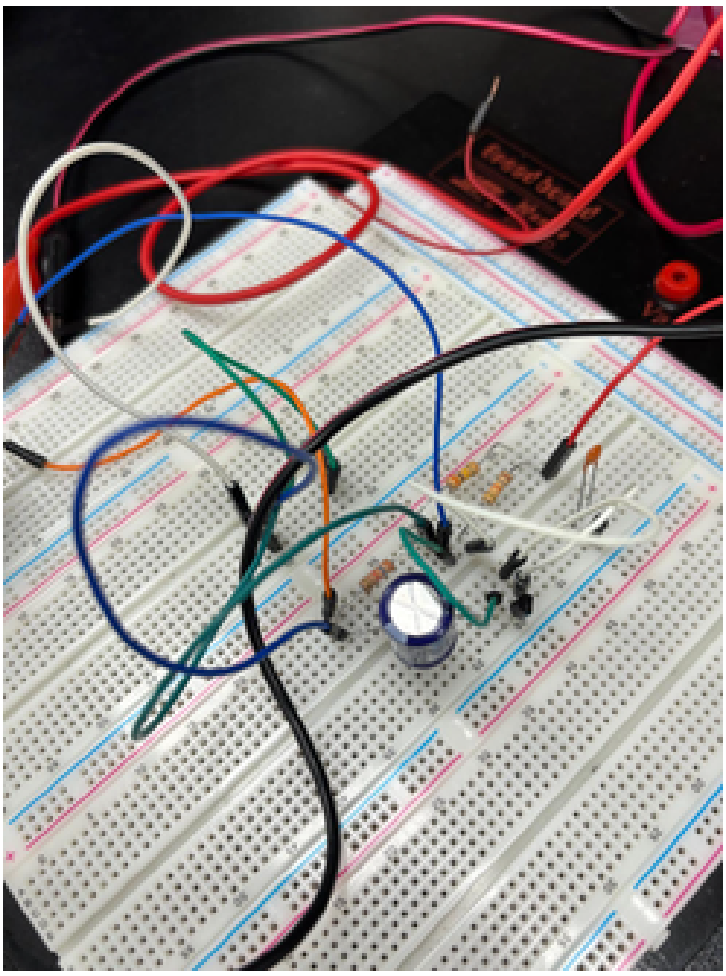
Common Base Amplifier						
α	β	g_m [S]	r_π [Ω]	A_I	A_V	A_V (V_{out}/V_{in})
0.992	130	0.0143	9140	131	0.955	



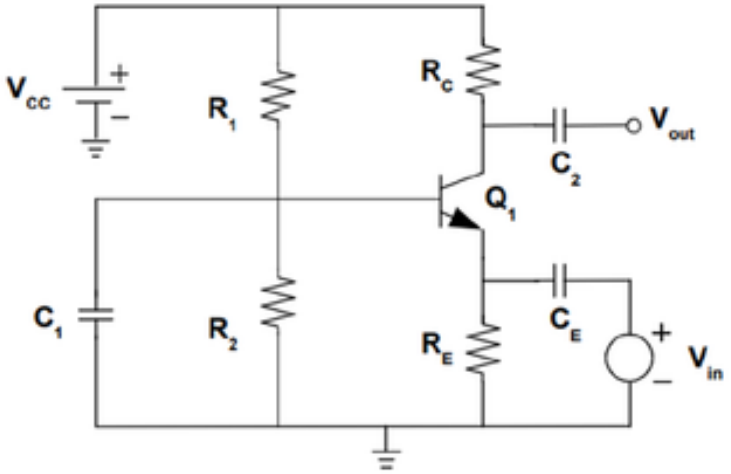
Lab Report:

Experiment 8.1:

1. The circuit is connected as shown in the figure.
2. Values are recorded.
3. Then connect the function generator to Vin as Voff = 0V, frequency as 1kHz.
4. Then decrease the Vpp from 0.25V to 0.15V by 0.01V and find the maximum Vpp of Vin to satisfy Vc has an undistorted waveform using oscilloscope.
5. Then calculate the maximum maximum Vpp for the amplifier.



Circuit:



Values:

Common Base Amplifier					
VB [V]	VC [V]	VE [mV]	IB [uA]	IC [uA]	I1 [uA]
1.20	7.049	0.595	1.28	396	37.38

Common Base Amplifier						
alpha	beta	gm	rπ [Ohm]	Ai	Av [gmRc]	Av [Vout/Vin]
1.059	309.37	0.01523	20315	0.944	274.14	40





VB value. VC value



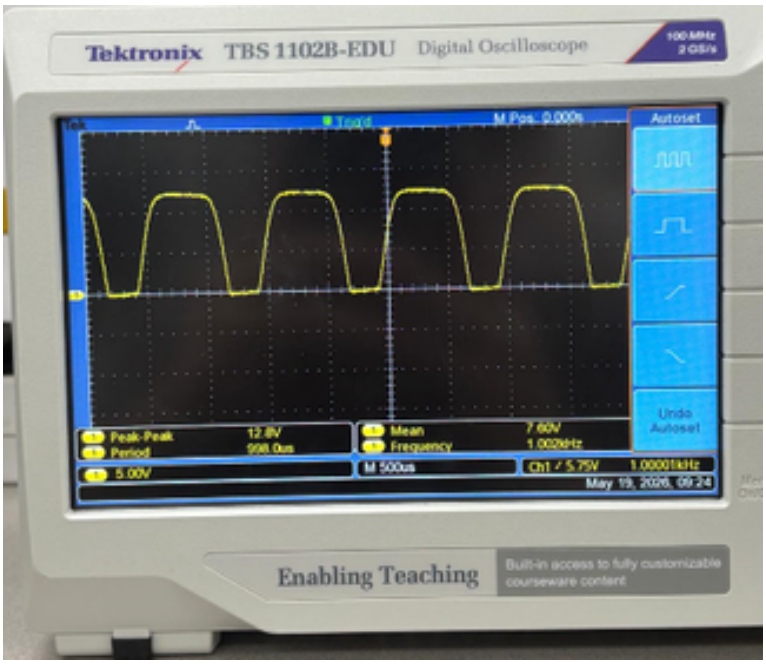


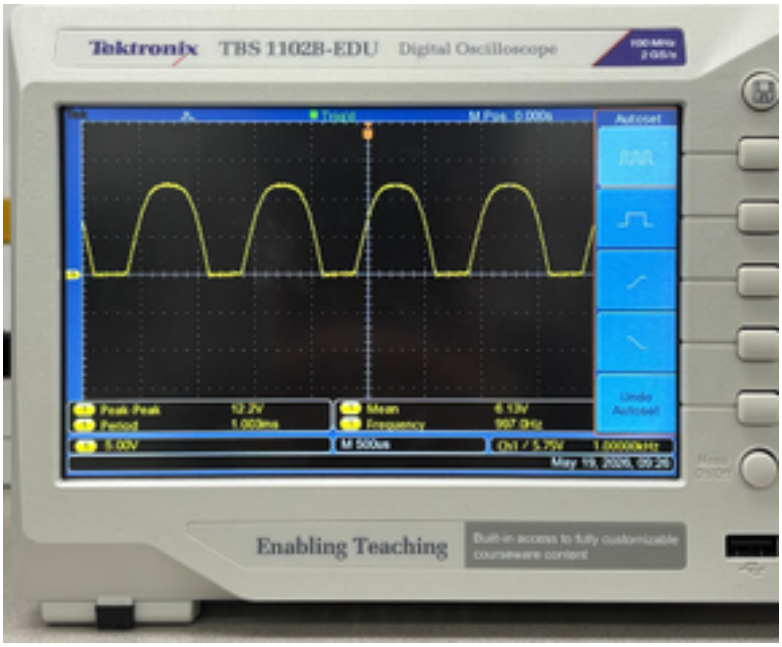
VE value.IC Value

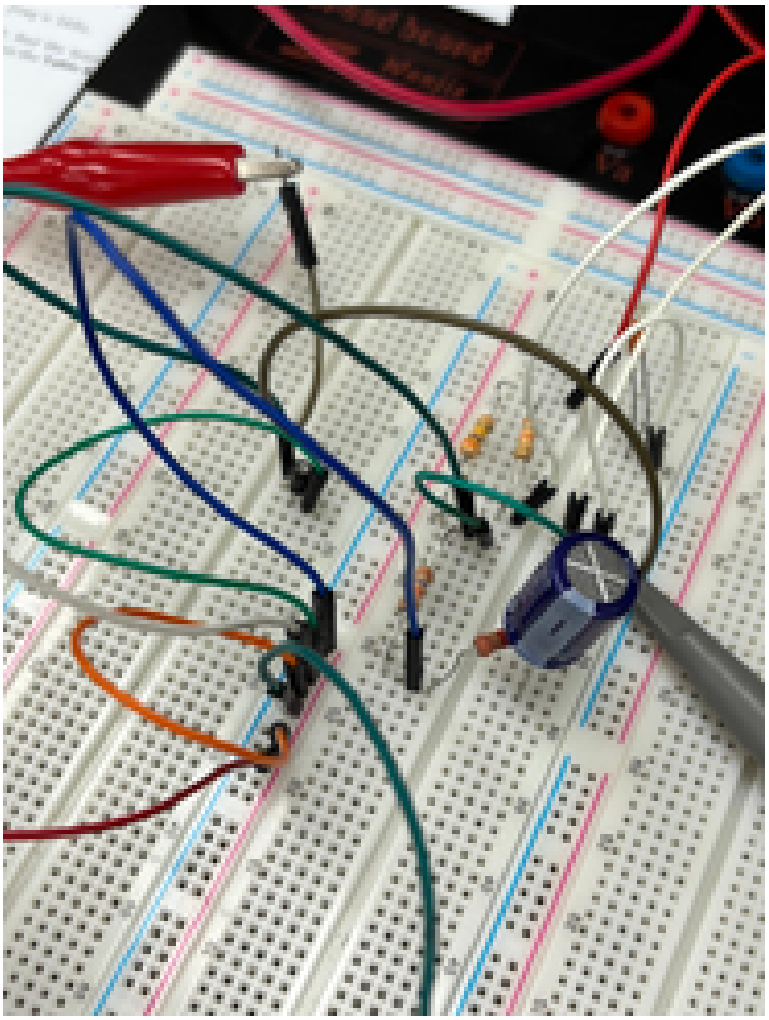


IB Value

Oscilloscope figure:

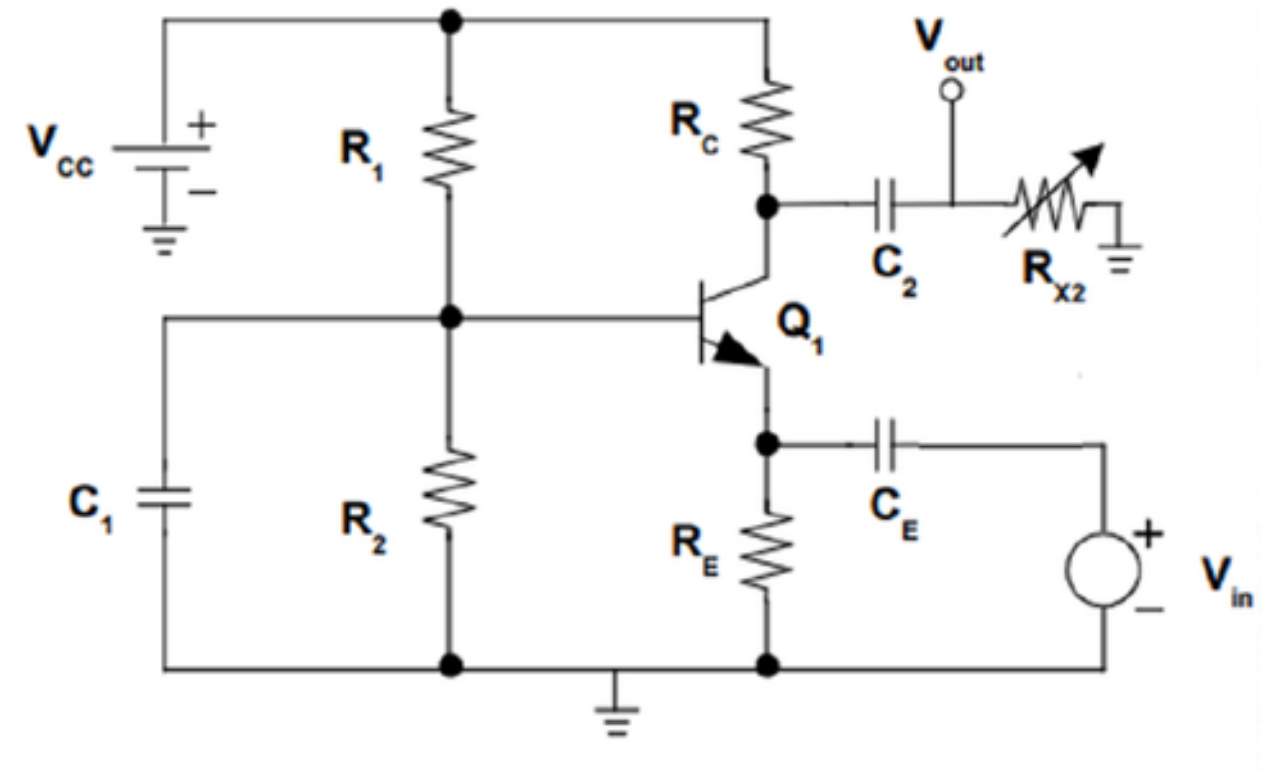

Tektronix
TBS 1102B-EDU
Digital Oscilloscope
Autoset
Undo
Autoset
Peak-Peak
12.8V
Period
998.0us
Mean
7.60V
Frequency
1.002kHz
5.00V
M 500us
Ch1 / 5.75V
1.00001kHz
May 19, 2026, 09:24
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Built-in access to fully customizable courseware content


Tektronix
TBS 1102B-EDU
Digital Oscilloscope
Autoset
Undo
Autoset
Peak-Peak
12.2V
Period
1.003ms
Mean
6.13V
Frequency
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5.00V
M 500us
Ch1 / 5.75V
May 19, 2026, 09:28
Enabling Teaching
Built-in access to fully customizable courseware content

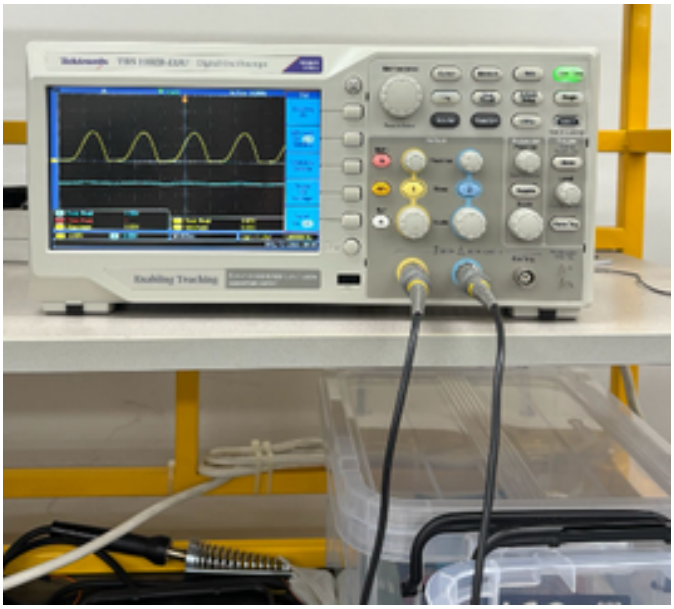


Experiment 8.2

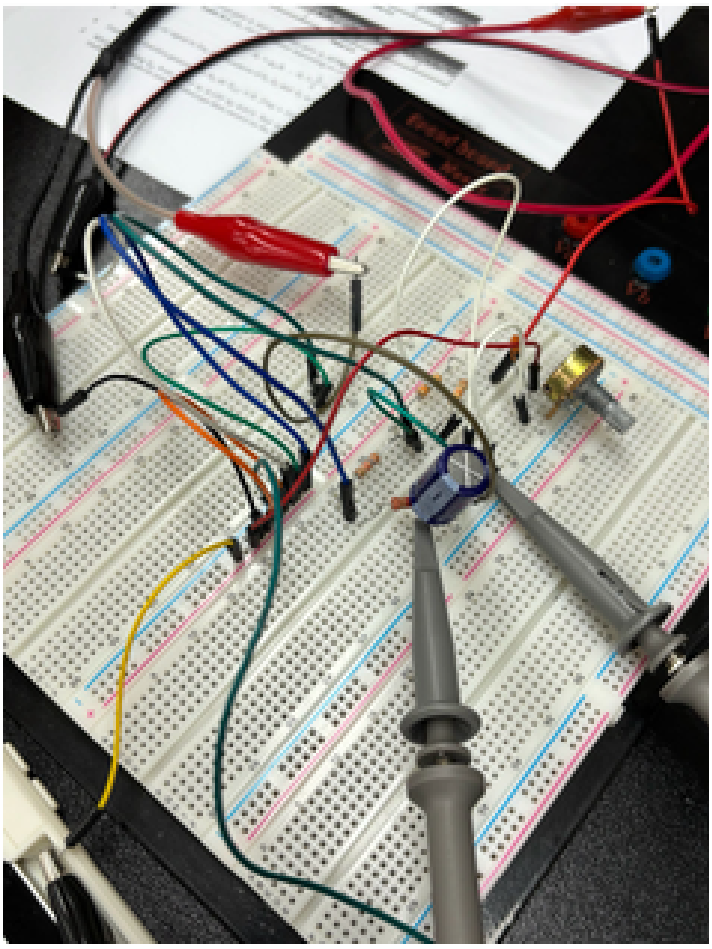
Circuit figure:



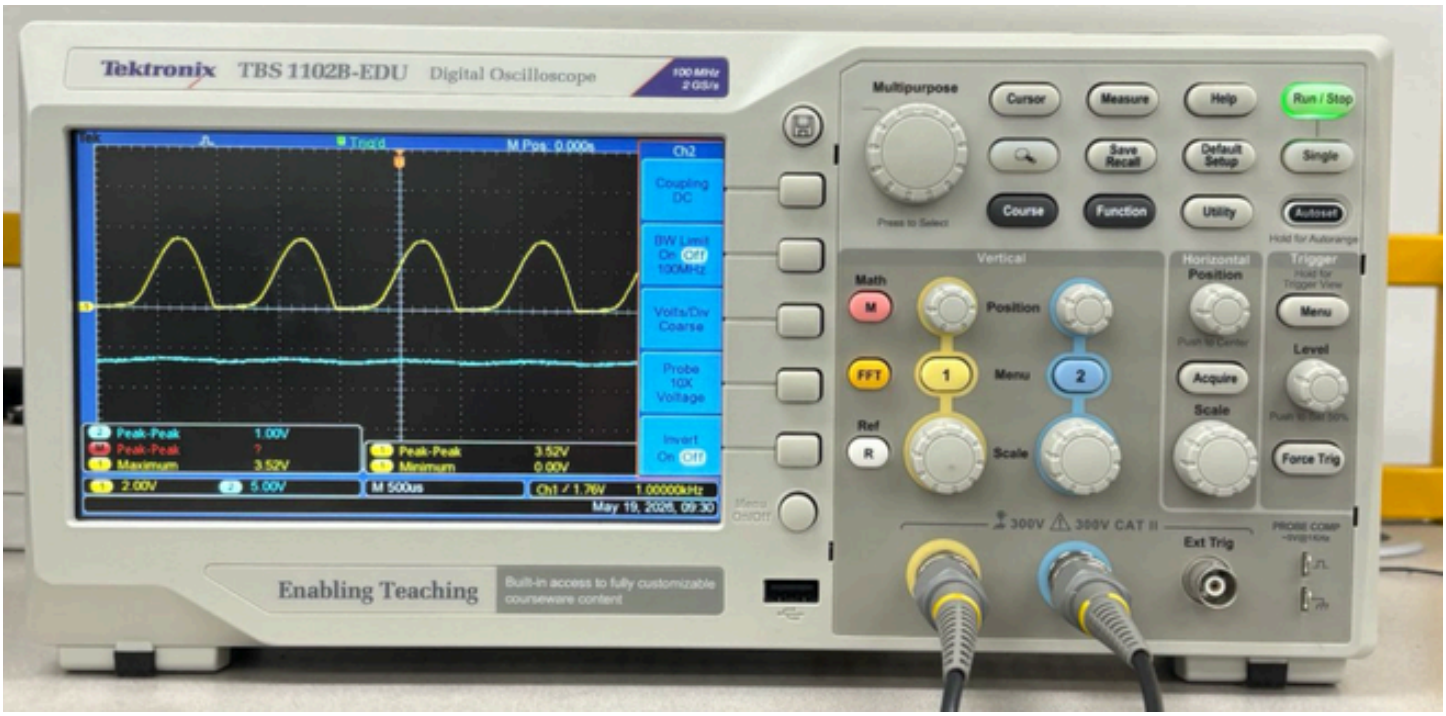
1. We then set the function generator to Vin as Voff =0, Vpp =0.1V and frequency as 1kHz.
2. We then record the Vpp value.



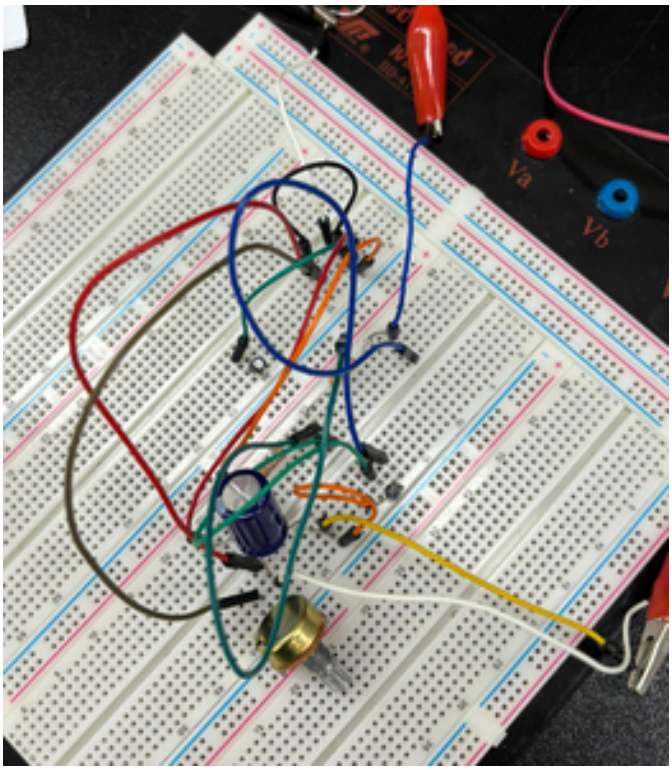
Circuit: Results:

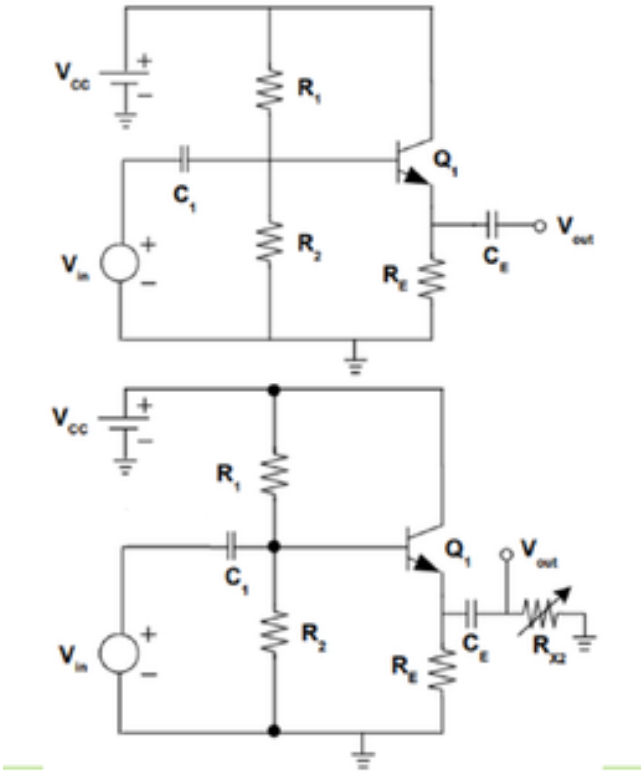


Common Base Amplifier		
Calculated		Measured
Rin	Rout	Rout
72Ohm	18kOhm	16.8kOhm



Experiment 8.3: Calculate the parameters and Output Impedance of EF



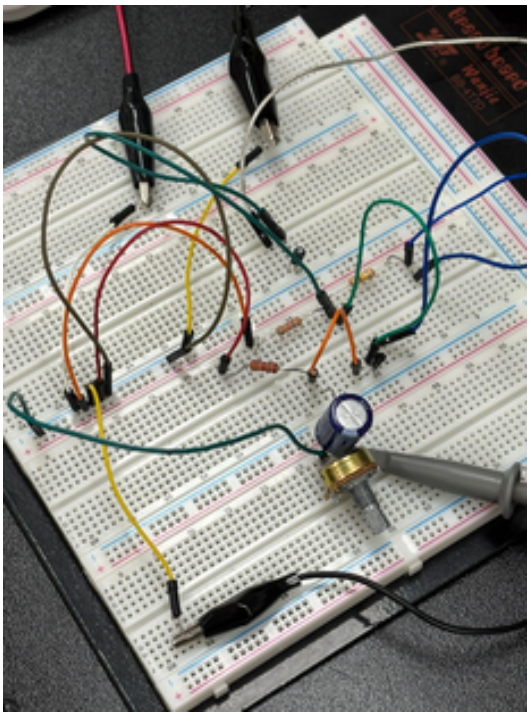


1. Connect the circuit as the figure shown above.
2. Using multimeter, measure the following values and obtain the remaining important metrics.
3. The steps are repeated by setting function generator to Vin as Voff= 0V and Vpp=0.1V
4. Then record the Vpp and the oscilloscope graph.

Emitter follower					
VB [V]	VC [V]	VE [mV]	IB [uA]	IC [uA]	I1 [uA]
1.20	12	601	1.26	369.74	37.1

Emitter follower						
alpha	beta	gm	rπ [Ohm]	Ai	Av [calculated]	Av [Vout/Vin]
0.997	293.4	0.0142	20.6k	294.4	0.955	0.95


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GW INSTEK GDM-8341
12076.
5 V


GW INSTEK GDM-8341
60010.
5 V


GW INSTEK GDM-8341
37140..
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